

ASSIGNMENT 3 | Linux and Git Version Control

Ajay Kumar | Roll Number: 22515990001 | GLA University | BCSE1522 | Semester 5

Project information, commits and branch changes

```
Last login: Fri Sep 11 14:00:46 on ttys000
ajaykumar@ajays-MacBook-Air ~ % cd ~/Ajay-GLA-LibrarySystem && {
  pwd
  cat library_info.txt
  git status
  git log --oneline -8
  git branch -a
  git remote -v
}
/Users/ajaykumar/Ajay-GLA-LibrarySystem
Name: Ajay Kumar
Roll Number: 22515990001
Subject: Application Build and Release Automation Lab
Project Name: Online Library Management System
Semester: 5th Semester
Project Objective: To develop an online library management system for managing books and library records.
Project Description: The system will help users manage book information, availability, and library operations efficiently.
On branch library-update
nothing to commit, working tree clean
8c8af63 (HEAD -> library-update, main, book-feature) Add book management module
0f59f06 Add project objective and description
466fe0f Add library project information
  book-feature
* library-update
  main
origin  git@github.com:ajay-cloudops/Ajay-GLA-LibrarySystem.git (fetch)
origin  git@github.com:ajay-cloudops/Ajay-GLA-LibrarySystem.git (push)
ajaykumar@ajays-MacBook-Air Ajay-GLA-LibrarySystem % nano library_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-LibrarySystem % git diff
git add library_info.txt
git commit -m "Add book issue and return to project scope"
diff --git a/library_info.txt b/library_info.txt
index 1406748..97ecd78 100644
--- a/library_info.txt
+++ b/library_info.txt
@@ -5,3 +5,4 @@ Project Name: Online Library Management System
  Semester: 5th Semester
  Project Objective: To develop an online library management system for managing books and library records.
  Project Description: The system will help users manage book information, availability, and library operations efficiently.
+Project Name: Online Library Management System with Book Issue and Return
[library-update 06a28cc] Add book issue and return to project scope
  1 file changed, 1 insertion(+)
ajaykumar@ajays-MacBook-Air Ajay-GLA-LibrarySystem % git switch main
nano library_info.txt
Switched to branch 'main'
ajaykumar@ajays-MacBook-Air Ajay-GLA-LibrarySystem % git add library_info.txt
git commit -m "Add book search to project scope"
[main 299e4fb] Add book search to project scope
  1 file changed, 1 insertion(+), 1 deletion(-)
ajaykumar@ajays-MacBook-Air Ajay-GLA-LibrarySystem % git merge library-update
git status
cat library_info.txt
Auto-merging library_info.txt
error: there was a problem with the editor 'vi'
Not committing merge; use 'git commit' to complete the merge.
On branch main
All conflicts fixed but you are still merging.
  (use "git commit" to conclude merge)

Changes to be committed:
```

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Merge conflict and manual resolution

```
Changes to be committed:
  modified:   library_info.txt

Name: Ajay Kumar
Roll Number: 22515990001
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Semester: 5th Semester
Project Objective: To develop an online library management system for managing books and library records.
Project Description: The system will help users manage book information, availability, and library operations efficiently.
Project Name: Online Library Management System with Book Issue and Return
[ajaykumar@ajays-MacBook-Air Ajay-GLA-LibrarySystem % git merge --abort ]
ajaykumar@ajays-MacBook-Air Ajay-GLA-LibrarySystem % git switch library-update ]
nano library_info.txt ]
Switched to branch 'library-update'
ajaykumar@ajays-MacBook-Air Ajay-GLA-LibrarySystem % git add library_info.txt
git commit -m "Correct project name in original line"
git switch main
git merge --no-edit library-update
git status
[cat library_info.txt ]
[library-update d227a9e] Correct project name in original line
 1 file changed, 1 insertion(+), 1 deletion(-)
Switched to branch 'main'
Auto-merging library_info.txt
CONFLICT (content): Merge conflict in library_info.txt
Automatic merge failed; fix conflicts and then commit the result.
On branch main
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   library_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
Name: Ajay Kumar
Roll Number: 22515990001
Subject: Application Build and Release Automation Lab
<<<<<< HEAD
Project Name: Online Library Management System with Book Search
=====
Project Name: Online Library Management System with Book Issue and Return
>>>>>> library-update
Semester: 5th Semester
Project Objective: To develop an online library management system for managing books and library records.
Project Description: The system will help users manage book information, availability, and library operations efficiently.
Project Name: Online Library Management System with Book Issue and Return
[ajaykumar@ajays-MacBook-Air Ajay-GLA-LibrarySystem % nano library_info.txt ]
ajaykumar@ajays-MacBook-Air Ajay-GLA-LibrarySystem % git diff --check ]
git add library_info.txt ]
git commit -m "Resolve merge conflict by combining book search, issue and return" ]
[main dd8a786] Resolve merge conflict by combining book search, issue and return
ajaykumar@ajays-MacBook-Air Ajay-GLA-LibrarySystem % cat library_info.txt
git status
git --no-pager log --oneline --graph --all
[git branch -a ]
Name: Ajay Kumar
```

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Final history, project structure and book module

```
Name: Ajay Kumar
Roll Number: 22515990001
Subject: Application Build and Release Automation Lab
Project Name: Online Library Management System with Book Search, Issue and Return
Semester: 5th Semester
Project Objective: To develop an online library management system for managing books and library records.
Project Description: The system will help users manage book information, availability, and library operations efficiently.
On branch main
nothing to commit, working tree clean
*   dd8a786 (HEAD -> main) Resolve merge conflict by combining book search, issue and return
| \
| * d227a9e (library-update) Correct project name in original line
| * 06a28cc Add book issue and return to project scope
* | 299e4fb Add book search to project scope
| /
* 8c8af63 (book-feature) Add book management module
* 0f59f06 Add project objective and description
* 466fe0f Add library project information
  book-feature
  library-update
* main
ajaykumar@ajays-MacBook-Air Ajay-GLA-LibrarySystem % mkdir -p ~/Desktop/Ajay-Assignment-3-Evidence

git --no-pager log --oneline --graph --all > ~/Desktop/Ajay-Assignment-3-Evidence/git-history.txt

git branch -a > ~/Desktop/Ajay-Assignment-3-Evidence/branches.txt

[git show --format=fuller --stat HEAD > ~/Desktop/Ajay-Assignment-3-Evidence/merge-details.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-LibrarySystem % find . -name .git -prune -o -print
cat src/book.txt
.
./config
./docs
./library_info.txt
./backup
./src
./src/book.txt
Book Management Module

Features:
1. Add new books to the library.
2. Search and view book details.
3. Update book information.
4. Check book availability.
ajaykumar@ajays-MacBook-Air Ajay-GLA-LibrarySystem % find "$HOME/Documents" "$HOME/Desktop" -type d -name "Cloud-3rd-year-26-27" -prune 2>/dev/null

if [ -d "$HOME/Cloud-3rd-year-26-27" ]; then
  echo "$HOME/Cloud-3rd-year-26-27"
fi
ajaykumar@ajays-MacBook-Air Ajay-GLA-LibrarySystem % cd ~/Documents
[git clone https://github.com/ajay-cloudops/Cloud-3rd-year-26-27.git
fatal: destination path 'Cloud-3rd-year-26-27' already exists and is not an empty directory.
ajaykumar@ajays-MacBook-Air Documents % cd ~/Documents/Cloud-3rd-year-26-27 && {
  mkdir -p Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem

  rsync -av --exclude='.git' \
    ~/Ajay-GLA-LibrarySystem/ \
    Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/

  mkdir -p Ajay-Kumar/Assignment-3/Evidence
```

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Submission files and Git history backup

```
mkdir -p Ajay-Kumar/Assignment-3/Evidence

cp -R ~/Desktop/Ajay-Assignment-3-Evidence/. \
  Ajay-Kumar/Assignment-3/Evidence/

touch Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/config/.gitkeep
touch Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/docs/.gitkeep
touch Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/backup/.gitkeep

git -C ~/Ajay-GLA-LibrarySystem bundle create \
  ~/Documents/Cloud-3rd-year-26-27/Ajay-Kumar/Assignment-3/Evidence/library-history.bundle \
  --all

git status --short
}
Transfer starting: 7 files
./
library_info.txt
backup/
config/
docs/
src/
src/book.txt

sent 984 bytes  received 94 bytes  347741 bytes/sec
total size is 586  speedup is 0.54
Enumerating objects: 22, done.
Counting objects: 100% (22/22), done.
Delta compression using up to 10 threads
Compressing objects: 100% (19/19), done.
Writing objects: 100% (22/22), 2.26 KiB | 2.26 MiB/s, done.
Total 22 (delta 6), reused 0 (delta 0), pack-reused 0 (from 0)
?? Ajay-Kumar/Assignment-3/
?? Ajay-Kumar/Practice-Exam-1/.DS_Store
?? Cloud-3rd-year-26-27/
[ajaykumar@ajays-MacBook-Air Cloud-3rd-year-26-27 % open ~/Documents/Cloud-3rd-year-26-27/Ajay-Kumar/Assignment-3/Evidence
ajaykumar@ajays-MacBook-Air Cloud-3rd-year-26-27 % cd ~/Documents/Cloud-3rd-year-26-27 && {
  git add Ajay-Kumar/Assignment-3
  git diff --cached --stat
  git remote -v
  git branch --show-current
}
Ajay-Kumar/Assignment-3/.DS_Store | Bin 0 -> 6148 bytes
Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/backup/.gitkeep | 0
Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/config/.gitkeep | 0
Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/docs/.gitkeep | 0
Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/library_info.txt | 7 ++++++
Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/src/book.txt | 7 ++++++
Ajay-Kumar/Assignment-3/Evidence/branches.txt | 3 +++
Ajay-Kumar/Assignment-3/Evidence/git-history.txt | 9 ++++++++
Ajay-Kumar/Assignment-3/Evidence/library-history.bundle | Bin 0 -> 2565 bytes
Ajay-Kumar/Assignment-3/Evidence/merge-details.txt | 11 ++++++++
10 files changed, 37 insertions(+)
origin https://github.com/ajay-cloudops/Cloud-3rd-year-26-27.git (fetch)
origin https://github.com/ajay-cloudops/Cloud-3rd-year-26-27.git (push)
upstream https://github.com/IshitaNainwal/Cloud-3rd-year-26-27.git (fetch)
upstream https://github.com/IshitaNainwal/Cloud-3rd-year-26-27.git (push)
practice-exam-2
ajaykumar@ajays-MacBook-Air Cloud-3rd-year-26-27 % git restore --staged Ajay-Kumar/Assignment-3/.DS_Store
```

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Assignment branch and successful GitHub push

```
origin https://github.com/ajay-cloudops/Cloud-3rd-year-26-27.git (push)
upstream https://github.com/IshitaNainwal/Cloud-3rd-year-26-27.git (fetch)
upstream https://github.com/IshitaNainwal/Cloud-3rd-year-26-27.git (push)
practice-exam-2
ajaykumar@ajays-MacBook-Air Cloud-3rd-year-26-27 % git restore --staged Ajay-Kumar/Assignment-3/.DS_Store
git fetch origin
git switch -c assignment-3 origin/main
A      Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/backup/.gitkeep
A      Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/config/.gitkeep
A      Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/docs/.gitkeep
A      Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/library_info.txt
A      Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/src/book.txt
A      Ajay-Kumar/Assignment-3/Evidence/branches.txt
A      Ajay-Kumar/Assignment-3/Evidence/git-history.txt
A      Ajay-Kumar/Assignment-3/Evidence/library-history.bundle
A      Ajay-Kumar/Assignment-3/Evidence/merge-details.txt
branch 'assignment-3' set up to track 'origin/main'.
Switched to a new branch 'assignment-3'
ajaykumar@ajays-MacBook-Air Cloud-3rd-year-26-27 % open Ajay-Kumar/Assignment-3/Evidence
ajaykumar@ajays-MacBook-Air Cloud-3rd-year-26-27 % git add Ajay-Kumar/Assignment-3
git restore --staged -- ':(glob)Ajay-Kumar/Assignment-3/**/.DS_Store'
git diff --cached --stat
git commit -m "Add Ajay Kumar Assignment 3 library system submission"
git push -u origin assignment-3
Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/backup/.gitkeep | 0
Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/config/.gitkeep | 0
Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/docs/.gitkeep | 0
Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/library_info.txt | 7 ++++++
Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/src/book.txt | 7 ++++++
Ajay-Kumar/Assignment-3/Evidence/branches.txt | 3 +++
Ajay-Kumar/Assignment-3/Evidence/git-history.txt | 9 ++++++++
Ajay-Kumar/Assignment-3/Evidence/library-history.bundle | Bin 0 -> 2565 bytes
Ajay-Kumar/Assignment-3/Evidence/merge-details.txt | 11 ++++++++
9 files changed, 37 insertions(+)
[assignment-3 50c48c3] Add Ajay Kumar Assignment 3 library system submission
9 files changed, 37 insertions(+)
create mode 100644 Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/backup/.gitkeep
create mode 100644 Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/config/.gitkeep
create mode 100644 Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/docs/.gitkeep
create mode 100644 Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/library_info.txt
create mode 100644 Ajay-Kumar/Assignment-3/Ajay-GLA-LibrarySystem/src/book.txt
create mode 100644 Ajay-Kumar/Assignment-3/Evidence/branches.txt
create mode 100644 Ajay-Kumar/Assignment-3/Evidence/git-history.txt
create mode 100644 Ajay-Kumar/Assignment-3/Evidence/library-history.bundle
create mode 100644 Ajay-Kumar/Assignment-3/Evidence/merge-details.txt
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 10 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (15/15), 4.08 KiB | 4.08 MiB/s, done.
Total 15 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'assignment-3' on GitHub by visiting:
remote: https://github.com/ajay-cloudops/Cloud-3rd-year-26-27/pull/new/assignment-3
remote:
To https://github.com/ajay-cloudops/Cloud-3rd-year-26-27.git
 * [new branch] assignment-3 -> assignment-3
branch 'assignment-3' set up to track 'origin/assignment-3'.
ajaykumar@ajays-MacBook-Air Cloud-3rd-year-26-27 % █
```